SECURE CODING REVIEW REPORT

Introduction

Secure coding is the practice of developing software that protects against vulnerabilities and cyber threats. This review analyzes a sample login application to identify security weaknesses and recommend improvements.

Application Reviewed

Python Login System

Vulnerabilities Identified

1. Hardcoded Passwords
   Passwords are stored directly in the source code.

Risk:
Attackers can easily access credentials.

2. Weak Input Validation
   User input is not properly validated.

Risk:
The application may be vulnerable to injection attacks.

3. Plain Text Password Storage
   Passwords are stored without encryption or hashing.

Risk:
Passwords can be stolen if the system is compromised.

Recommendations

• Use bcrypt for password hashing.
• Validate all user inputs.
• Avoid hardcoded credentials.
• Implement secure authentication mechanisms.
• Follow OWASP Secure Coding Guidelines.

Conclusion

Secure coding practices help prevent vulnerabilities and improve software security. Implementing the recommended measures will strengthen the application's security posture.